Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2019
EE 3215 – Broadband Communication

Instructions for Candidates:
Answer **any Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 03
Time : **03 hours**

Q1. (i) Define DSL? (3 marks)

(ii) Briefly explain the broadband communication and discuss the reasons to get low speed Broadband? (5 marks)

(iii) Explain the following terms (6 marks)

(a)Baud rate

(b)Bit rate

(iv) Distinguish between ADSL and VDSL? (6 marks)

Q2. (i) What is meant by signal to noise ratio? (3 marks)

(ii)What is loading coil? (3 marks)

(iii) Briefly Explain Quadrature Amplitude Modulation (QAM) (3 marks)

(iv) Explain the function of Filter/splitter, ADSL Modem/Router, and Brass in an ADSL Network? (5 marks)

(v) Define PSTN and how voice signals are carried over the public switched telephone Networks? (6 marks)

Q3. (i) Briefly explain the DMT? (3 marks)

(ii)Explain the Broadband ISDN? (5 marks)

(iii) What are the functional requirements have to addressed by a telecommunication Company to enable a mass market for ADSL? (6 marks)

(iv) Discuss the Business opportunity for ADSL Bases Broadband Network? (6 marks)

Q4. (i) Discuss the principle of OFDM modulation scheme? (2 marks)

(ii) Differentiate Cellular telephony and Cordless telephony.? (3 marks)

(iii)) Why uplink frequency is different from downlink frequency in satellite. Explain? (4 marks)

(iii) What are the advantages of BPL over other connection? (4 marks)

(v)Draw the architecture of the BPL and briefly explain the functions of repeater and Extractor in BPL? (7 marks)

Q5. (i) What is meant by BRAN? (3 marks)

(ii)Write description about DSLAM? (5 marks)

(iii)Draw the ATM protocol architecture and briefly explain the functions of physical Medium- Dependent (PMD) sub layer and the transmission convergence (TC) sub layer? (6 marks)

(iv) What are the two primary types of interfaces in ATM switches and explain these two Primary types of interfaces? (6 marks)

Q6. (i) What is the need for WIMAX? (2 marks)

(ii) List any four salient features supported by wi-max? (4 marks)

(iii) Briefly explain TDD and FDD? Distinguish between TDD-LTE and FDD-LTE? (6 marks)

(iv)Write the short notes on below wireless broadband technologies? (8 marks)

a). HSPA

b). Wi-Fi